

A Few Definitions and Theorems

Definition. A sequence $\{a_n\}$ is **increasing** if $a_{n+1} \geq a_n$ for all n .

Definition. A sequence $\{a_n\}$ is **strictly increasing** if $a_{n+1} > a_n$ for all n .

Definition. A sequence $\{a_n\}$ is **decreasing** if $a_{n+1} \leq a_n$ for all n .

Definition. A sequence $\{a_n\}$ is **strictly decreasing** if $a_{n+1} < a_n$ for all n .

Definition. A sequence $\{a_n\}$ is **monotone** if it is either increasing or decreasing.

Definition. A sequence $\{a_n\}$ is **bounded above** by M if $a_n \leq M$ for all n .

Definition. A sequence $\{a_n\}$ is **bounded below** by m if $a_n \geq m$ for all n .

Definition. A sequence $\{a_n\}$ is **bounded** if it is *both* bounded above and bounded below.

Theorem. If a sequence is increasing and bounded above, then it must converge.

Theorem. If a sequence is decreasing and bounded below, then it must converge.

Bounded Monotone Convergence Theorem. A monotone sequence converges if and only if it is bounded.

Theorem. If r is a fixed number such that $|r| < 1$, then $\lim_{n \rightarrow \infty} r^n = 0$.